Version

2.1

MORPHO SOFTWARE, INC.

MorphoBlock Library Module

Technical Reference

MORPHOBLOCK™ LIBRARY MODULE

Technical Reference

© Morpho Software, Inc.
3944 State Street • Suite 340
Santa Barbara, CA 93105, USA
Phone 805.569.9022 • Fax 805.569.5382

Legal Statement

Please read the following Terms of Use carefully. By using this document and accompanying software (collectively, "Library"), you acknowledge that you have read the following Terms of Use and that you accept and will be bound by them, as they may be modified by Morpho Software, Inc., from time to time, as printed in updates to this document, or available by request from Morpho Software, Inc. If you do not agree to these Terms of Use, do not use this Library. Any use of the Library is at the sole risk of the user.

Legal Name and Address

Morpho Software, Inc., is incorporated in California with headquarters located at 3944 State Street, Suite 340, Santa Barbara, California 93105. Our phone number is 805-569-9022. For the purposes of these Terms of Use, references to Morpho Software, shall include any affiliates, directors, officers, or employees.

Information Copyright 2001, 2002 by Morpho Software, Inc. All information in the Library is protected by copyright in the United States of America and in other countries. Unless otherwise specified (see the Limited Disclaimer of Copyright below), no one has permission to copy or republish, in any form, any information found on this Web site.

Limited Disclaimer of Copyright

Authorized persons are hereby permitted to view the information available from this Library for informational and evaluation purposes only. No part of the information in this Library can be otherwise redistributed, copied, or reproduced in any form without prior written consent of Morpho Software, Inc.

Disclaimer of Warranties

This information is provided "AS IS" without warranty of any kind, either expressed or implied, including, but not limited to, the implied warranties of merchantability, fitness for a particular purpose, or non-infringement.

MORPHO SOFTWARE, INC., MAKES NO REPRESENTATION AS TO THE ACCURACY, TIMELINESS, OR COMPLETENESS OF THE LIBRARY MATERIALS. MORPHO SOFTWARE, INC., PERIODICALLY AMENDS, CHANGES, ADDS, DELETES, UPDATES, OR ALTERS THE INFORMATION, INCLUDING, WITHOUT LIMITATION, THE TERMS OF USE OF THE LIBRARY, WITHOUT NOTICE. FURTHER, MORPHO SOFTWARE, INC., ASSUMES NO LIABILITY OR RESPONSIBILITY FOR ANY ERRORS OR OMISSIONS IN THE CONTENT OF THE LIBRARY. MORPHO SOFTWARE, INC. SPECIFICALLY DISCLAIMS ANY DUTY TO UPDATE THE INFORMATION IN THE LIBRARY.

REFERENCE TO THE CORPORATION AND ITS LIBRARY ARE PROVIDED “AS IS” WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED. IN NO EVENT SHALL MORPHO SOFTWARE, INC., OR ANY OTHER PARTY INVOLVED IN THE CREATION, PRODUCTION, OR DELIVERY OF THE INFORMATION IN THIS LIBRARY BE LIABLE FOR ANY SPECIAL, INCIDENTAL, INDIRECT, OR CONSEQUENTIAL DAMAGES OF ANY KIND, OR ANY DAMAGES WHATSOEVER RESULTING FROM LOSS OF USE, COMPUTER VIRUS OR

SYSTEM FAILURE, DATA OR PROFITS, WHETHER OR NOT ADVISED OF THE POSSIBILITY OF DAMAGES, AND ON ANY THEORY OF LIABILITY, ARISING OUT OF OR IN CONNECTION WITH THE USE OR PERFORMANCE OF THIS LIBRARY (OR THE INFORMATION OR SERVICES PROVIDED WITH THIS LIBRARY), OR USERS' INABILITY TO USE THE INFORMATION OR MATERIALS CONTAINED IN THE LIBRARY (OR ANY OTHER LIBRARY). IN NO EVENT SHALL MORPHO SOFTWARE, INC.'S LIABILITY TO USERS FOR ALL DAMAGES, LOSSES, AND CAUSES OF ACTION (WHETHER IN CONTRACT OR TORT, INCLUDING, BUT NOT LIMITED TO, NEGLIGENCE OR OTHERWISE) EXCEED THE AMOUNT, IF ANY, PAID BY USERS TO MORPHO SOFTWARE, INC., FOR ACCESSING THIS LIBRARY. THE USER SHOULD CONSULT HIS OR HER OWN LEGAL ADVISORS REGARDING THE APPLICABILITY OF STATE STATUTES REGARDING LIMITATIONS ON LIABILITY.

DESCRIPTIONS OF, OR REFERENCES TO, PRODUCTS, SERVICES, OR PUBLICATIONS WITHIN THE MORPHO SOFTWARE, INC., LIBRARY THAT ARE NOT PRODUCED AND OWNED SOLELY BY MORPHO SOFTWARE, INC., DO NOT IMPLY ENDORSEMENT OF THAT PRODUCT, SERVICE, OR PUBLICATION. MORPHO SOFTWARE, INC. MAKES NO WARRANTY OF ANY KIND WITH RESPECT TO THE SUBJECT MATTER INCLUDED HEREIN OR THE PRODUCTS LISTED HEREIN, THAT ARE NOT PRODUCED AND OWNED BY MORPHO SOFTWARE, INC. MORPHO SOFTWARE, INC. SPECIFICALLY DISCLAIMS ALL WARRANTIES, EXPRESSED, IMPLIED, OR OTHERWISE, INCLUDING, WITHOUT LIMITATION, ALL WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, FOR ALL PRODUCTS, SERVICES, OR PUBLICATIONS NOT PRODUCED AND OWNED BY MORPHO SOFTWARE, INC.

SOME STATES DO NOT ALLOW THE DISCLAIMER OF WARRANTIES OR THE EXCLUSION OF LIABILITY FOR CONSEQUENTIAL DAMAGES, SO THE ABOVE LIMITATIONS MAY NOT APPLY TO YOU IN ALL CASES.

Indemnification by User

USER AGREES TO INDEMNIFY, DEFEND, AND HOLD MORPHO SOFTWARE, INC., HARMLESS FROM ANY LIABILITY, LOSS, CLAIM, AND EXPENSE, INCLUDING ATTORNEY'S FEES RELATED TO A USER'S VIOLATION OF THESE TERMS OF USE OR THE USE OF THE SERVICES AND INFORMATION PROVIDED IN THIS LIBRARY.

Typographical Errors

This Library could include technical inaccuracies or typographical errors.

Submission of Information

All information transmitted or otherwise submitted to or via this Library shall be deemed to be the property of Morpho Software, Inc. and Morpho Software, Inc. shall be free to use such information for any lawful purpose. Morpho Software, Inc. shall not have any obligation to keep such information confidential except as otherwise specifically agreed in writing by Morpho Software, Inc. or as required by law. Users should note that electronic transmissions associated with this Library are not confidential and may be read or intercepted by others.

Table of Contents

INTRODUCTION **3**

Supporting Configuration **3**
- Hardware Requirements 3
- Software Requirements – Linux Configuration 3
- Software Requirements – Windows Configuration 4

Packaging Information **4**
- Common files 4
- Linux Specific Files 4
- Windows Specific Files 5

Installation **5**
- Installation on Linux Machines 5
- Installation on Windows Machines 5

APIS **6**

Listing of APIs **6**
- Initialize and Update Functions 6
- Query Functions 6
- Error Reporting Function 6

API Reference **6**
- mb_cleanup 7
- mb_init 8
- mb_genact 10
- mb_update 11
- mb_getinfo 13
- mb_query 14

CONFIGURATION SETTINGS **15**

Sample Configuration **15**

Must-Have Settings **16**
- CertPath 16
- MLPath 16
- DLPath 16
- LLPath 16

Optional Settings **16**
- ALPath 16
- TableSize 16
- ALSize 16

Local Administrator Override **17**

Chapter 1

Introduction

MorphoBlock is a comprehensive software solution that raises the bar in the field of objectionable content filtering. The MorphoBlock library module serves as a high-performance URL block / pass list, and the lists can be updated on the fly without disturbing normal operations. In addition, to combat the ever-changing list of sites that must be filtered, each MorphoBlock server updates its database of blocked sites on a daily basis from our centralized, secure data center, increasing performance.

Supporting Configuration

MorphoBlock library module currently supports Linux and Windows Operating Systems running on the Intel x86 –based machines. Following are the system requirements for using and running MorphoBlock library module.

Hardware Requirements

In order to get reasonable response time from queries, minimum requirement of physical memory at run-time is 140MB. Suggested configuration is to reserve 256MB of physical memory for MorphoBlock. (Thus you may need 512MB or more physical memory for the OS, the MorphoBlock, and your application.)

The minimum CPU requirement is Intel Pentium II 266. However, such a low-end CPU may delay the loading time and query response a lot. Our suggestion is to use the Pentium 4 1.8GHz class CPU, or dual Pentium III 1G configuration.

The minimum requirement of disk space is 20GB. Reserving 40GB is suggested.

Software Requirements – Linux Configuration

The Linux Kernel version should be 2.2 or above. The g++ version should be 2.95.3 or above, with libstdc++ 2.9 or above installed. If your application is written in pure C, please contact Morpho Software for further support information.

Software Requirements – Windows Configuration

Only Windows 2000 / XP are supported now, and we suggest XP. Our lab test shows that MorphoBlock running on Windows XP performs much better than running on Windows 2000. Microsoft Visual C++ 6 (with SP5) or Microsoft VisualStudio.Net is required. The Microsoft VisualStudio.Net is suggested since its C++ library is much better than the one with Visual C++ 6.

Packaging Information

MorphoBlock library modules are shipped in the shared library form. Following are the manifest of the package.

Common files

`morf.h`	API declaration header file
`morf_msg.h`	Message definition header file required by `morf.h`
`mb2.cfg`	Sample configuration file
`morpho.cert`	Certificate file
`sitelist.xml`	Master list file
`detlist2.xml`	Sample update list file
`override.txt`	Sample administrator override file
`qrybench.cpp`	Sample program: Query benchmark
`list`	Data file used by `qrybench.cpp`

Linux Specific Files

`libmorpho.so`	Shared library file
`Makefile`	Sample Makefile for `qrybench.cpp`

Windows Specific Files

`mb2.lib`	Library file for linking
`mb2.dll`	Dynamic Linked Library for Windows applications

Installation

Installation on Linux Machines

Linux package of MorphoBlock library comes in gzipped or bzipped tarball format per your request. For the gzipped version, simply use `tar` to install MorphoBlock library into desired directory: The following example demonstrates how to install the library into `/home/someone/MorphoBlock`:

```
tar zxf MorphoBlock.tar.gz /home/someone/MorphoBlock
```

For bzipped tarball, you need to figure out if your `tar` supports `j` command. If so, following example shows you how to install into `/home/someone/MorphoBlock`:

```
tar jxf MorphoBlock.tar.bz2 /home/someone
```

If your `tar` does not support `j` command, try this:

```
bzcat MorphoBlock.tar.bz2 | tar x /home/someone
```

Installation on Windows Machines

Windows package comes in self-extracted zip-executable format. Simply run mb2.exe, choose the directory you want to install, then you are all set.

Chapter 2

APIs

MorphoBlock APIs are defined in the file `morf.h`. The return values, error codes, and error messages of these APIs are defined in the file `morf_msg.h`. Please note that **MorphoBlock 2.1 APIs for Linux version are not thread safe**, as we do not make assumption on which thread library implementation being used. If you need to have thread-safe APIs, please contact us directly.

Listing of APIs

Initialize and Update Functions

- `int mb_init(const char* pszConfigFile);`
- `int mb_cleanup();`
- `int mb_update(const char* pszConfigFile);`
- `int mb_genact();`

Query Functions

- `int mb_getinfo(TMBInfo *);`
- `TTag mb_query(const char* pszUrl);`

Error Reporting Function

- `int mb_geterr(char *);`

API Reference

The following is the detailed API reference.

mb_cleanup

NAME

```
int mb_cleanup
```

SYNOPSIS

```
#include "morf.h"

int mb_cleanup();
```

DESCRIPTION

The `mb_cleanup()` function returns all system resources allocated by MorphoBlock back to the operating system.

RETURN VALUES

`MB_OKAY`

ERRORS

This function does not return any error code.

SEE ALSO

`mb_init`

mb_init

NAME

`int mb_init`

SYNOPSIS

```
#include "morf.h"

int mb_init(const char *pszConfigFile);
```

DESCRIPTION

The `mb_init()` function opens up the configuration file, whose full path is specified in `pszConfigFile`. Then it allocates memory, loads data from main database file specified in configuration settings, and initializes internal filter database to prepare MorphoBlock for receiving URL queries.

The `mb_init()` function returns `MB_OKAY` even if there is an invalid setting of hash table size in the configuration file. However, it will ignore this invalid value and use the default value instead.

If there are not sufficient system resources for `mb_init()` to utilize, it will return `MB_ERROR` and the error code you get from `mb_geterr()` is still `MB_ERROR`, too. You will not be able to do any query unless `mb_init()` is successfully completed.

You may experience some waiting for the completion of `mb_init()`. Since the database size will eventually grow to millions, we suggest that if you need to specify a timeout value for `mb_init()`, this value should not be lower than 180 seconds for a database containing one million of entries.

RETURN VALUES

`MB_OKAY, MB_ERROR`

ERRORS

`MB_ECIPHER`	Invalid cipher or public key
`MB_EFILEDUPENTRIES`	Duplicated file entries
`MB_EFILEFORMAT`	File format error
`MB_EFILENOTFOUND`	File not found

MB_EFILESETTINGERR	File setting error
MB_EFILEVERSION	File version mismatch
MB_EOPEN	Fails to open the configuration file
MB_ESTAMPMISMATCH	Delta list stamp mismatch
MB_WTABLESIZE	Invalid hash table size

SEE ALSO

mb_cleanup, mb_update, mb_geterr

mb_genact

NAME

`int mb_genact`

SYNOPSIS

```
#include "morf.h"

int mb_genact();
```

DESCRIPTION

The `mb_genact()` function generates an active list for off-line processing, which identifies the objectionable sites on the active list.

RETURN VALUES

`MB_OKAY, MB_ERROR`

ERRORS

`MB_EGENACTLIST`	fails to generate active list

SEE ALSO

`mb_update, mb_geterr`

mb_update

NAME

`int mb_update`

SYNOPSIS

```
#include "morf.h"

int mb_update(const char* pszConfigFile);
```

DESCRIPTION

The `mb_update` function opens up the configuration file, whose full path is specified in `pszConfigFile`. Then it loads the updated data from the data file specified in the configuration settings, and updates the internal filter database of MorphoBlock using these data.

If you change the hash table size setting in the configuration file, you need to call `mb_cleanup()` and `mb_init()` to reflect your settings. The `mb_update()` reads the configuration file only for referencing the update data file path. However, it does parse the configuration file, and it will return `MB_ERROR` without updating the filter database if it detects any error inside the configuration file except for invalid hash table size.

If there are not sufficient system resources for `mb_update()` to utilize, it will return `MB_ERROR` and the error code you get from `mb_geterr()` is still `MB_ERROR`, too. Internal database may be partially updated, and the MorphoBlock will keep running. The partial update problem will be fixed automatically after next successful completion of `mb_update()`.

RETURN VALUES

`MB_OKAY, MB_ERROR`

ERRORS

`MB_ECIPHER`	Invalid cipher or public key
`MB_EFILEDUPENTRIES`	Duplicated file entries
`MB_EFILEFORMAT`	File format error
`MB_EFILENOTFOUND`	File not found

MB_EFILESETTINGERR	File setting error
MB_EFILEVERSION	File version mismatch
MB_EOPEN	Fails to open the configuration file
MB_ESTAMPMISMATCH	Delta list stamp mismatch
MB_WTABLESIZE	Invalid hash table size

SEE ALSO

mb_cleanup, mb_init, mb_geterr

mb_getinfo

NAME

`int mb_getinfo`

SYNOPSIS

```
#include "morf.h"

int mb_getinfo(TMBInfo *);
```

DESCRIPTION

The `mb_getinfo` function returns MorphoBlock system information including library version number, main database version, database update stamp, and the number of entries inside filter database.

The `TMBInfo` structure is defined in `morf.h`, please be sure to allocate a valid instance for the pointer before you pass it in.

RETURN VALUES

`MB_OKAY, MB_ERROR`

ERRORS

`MB_ELISTEMPTY`	Database is empty or not initialized

mb_query

NAME

```
int mb_query
```

SYNOPSIS

```
#include "morf.h"

TTag mb_query(const char* pszUrl);
```

DESCRIPTION

The `mb_query` function returns the state of the specified URL. The specified URL is assumed to be syntax-error free. For compatibility with prior versions, you can also pass in domain names only, but it's deprecated and may obsolete in future versions.

RETURN VALUES

The return values is `MB_SPASSED` or the Boolean-OR combination of `MB_SBLOCKED` and at least one of the following:

`MB_SPORN`

`MB_SHATE`

`MB_SVIOLANCE`

`MB_SGAMBLE`

ERRORS

There is no error code and message this API will raise.

Chapter

3

Configuration Settings

MorphoBlock library module utilizes external configuration settings in order to provide you more flexibility. These settings are stored in a plain text file, and you can use your favorite text editor to modify its contents. However, most changes of configuration settings will be reflected only after restarting MorphoBlock. Please be careful of configuration changes.

Sample Configuration

Here are the contents of the sample configuration file, `mb2.cfg`.

```
# MorphoBlock Configuration File

# Following four properties MUST be set: CertPath, MLPath, DLPath, LLPath
# CertPath is the certificate issued by Morpho Software, this file MUST exist
CertPath =      morpho.cert

# MLPath is the main list file, this file MUST exist
MLPath          =      sitelist.xml

# DLPath is the patch list issued by Morpho Software
DLPath          =      detlist.xml

# LLPath is the Administrator override list
LLPath          =      adminlist.txt

# ALPath is the active list file generated by MorphoBlock
# empty to disable this functionality
ALPath          =      active.lst

# Main hash table size, should be a prime number
# empty to use default value 5642083
TableSize=

# Active list table size, should be a prime number
# empty to use default value 41263
ALSize          =
```

We will go through every property settings in following sections.

Must-Have Settings

The settings introduced in this section are the "must-have" settings. You have to designate values for these properties **no matter the values are invalid or valid**. Empty value settings will be treated as syntax errors and thus the whole configuration file will be treated as invalid.

CertPath

This property indicates the full path of certificate file issued by Morpho Software. **The specified file MUST exists while running MorphoBlock**.

MLPath

This property indicates the full path of master data file shipped with the MorphoBlock library module. **The specified file MUST exists while running MorphoBlock**.

DLPath

This property indicates the full path of database update file provided by Morpho Software. The specified file may not exist if there is no update for current master data file.

LLPath

This property indicates the full path of a text file containing administrator overrides settings. The specified file may not exist if there is no overrides provided by local administrator.

Optional Settings

The settings introduced in this section are the optional settings. You don't need to change the default settings unless you are suggested to.

ALPath

This property indicates the full path of where the active list file should be generated. The active list file is designed for off-line processing. If you empty this property, the `mb_genact()` function will be disabled and always return `MB_OKAY` without doing anything.

TableSize

This property let you set the hash table size of the internal filter database of MorphoBlock. This value should be a prime number and bigger than default value. If it is not a prime number, or smaller than default value, then the default value will be used instead of your manual setting.

ALSize

This property let you set the hash table size of the internal active list database of MorphoBlock. This value should be a prime number and bigger than default value. If it is not a prime number, or smaller than default value, then the default value will be used instead of your manual setting. The setting of ALSize will be valid if and only if the ALPath property has a valid setting.

Local Administrator Override

Local administrators can override the settings provided by filter database. The override settings should be stored in a plain text file in such format:

```
<url or site> <tags>

<url or site> <tags>

<url or site> <tags>
```

The <url or site> can be a syntax error-free URL string or domain name. No spaces, asterisks, backslashes or characters rather than US ASCII are allowed in the URL string. For example,

```
http://www.somewhere.com/okay/index.html BLOCKED PORN

google.com PASSED
```

are good settings. Following examples are bad settings,

```
http://www.somewhat.com/bad / *.html PASSED

.google.com PASSED
```

Lines containing bad URL or site names will be automatically ignored and thus ineffective.

For the <tags>, you need to specify that it is `PASSED` or `BLOCKED`. If it is `BLOCKED`, then you can specify its reason. Valid reasons are,

```
PORN

GAMBLE

VIOLENCE

HATE
```

Tags are not case sensitive. However, if the MorphoBlock detects there are semantic errors in tags, this line will be ignored automatically, too. For example,

```
http://www.badsettings.com/bad/index.html PASSED BLOCKED

http://www.badsettings.com/bad2/index.html PASSED PORN
```

MorphoBlock will automatically ignore these two lines.

NOTE

Although you specify exact URLs in the administrator override, MorphoBlock does not perform file-level blocking. Instead, it performs directory level blocking. For example, if you add following line in the administrator override file

http://www.somewhere.com/okay/index.html BLOCKED PORN

Every file under

http://www.somewhere.com/okay/

will be blocked.